

Visitors Counter

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Presented here is a simple counter that counts the number of visitors entering or exiting an auditorium or any other place where you have installed this circuit at the gates. On receiving an interrupt from light-dependent resistor (LDR) sensors, the circuit increments the count and shows it on a 7-segment display.

With these units installed at the entrance and exit gates, you can calculate the number of visitors present in the room by subtracting the count

at the exit gate from the count at the entrance gate. The system should be installed on a door such that only one person can cross through at a time, interrupting the light falling on the LDR sensor.

Circuit and working

The circuit is built around popular CD4026 counter ICs (IC1 and IC2), light-dependent resistor (LDR1), transistor BC547 (T1), common-cathode seven-segment displays (DIS1 and DIS2) and a few other components. The advantage of using CD4026 counter IC is that it drives a 7-segment display without the need of a driver IC.

The resistance of LDR1 decreases when the intensity of light falling on it increases and vice versa. In dark or absence of light, the LDR exhibits a resistance in the range of mega-ohms, which decreases to a few hundred

PARTS LIST

Semiconductors:

- IC1, IC2 - CD4026 counter
- T1 - BC547 npn transistor
- LED1 - 5mm red LED
- DIS1-DIS2 - LTS543 common-cathode 7-segment display

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-kilo-ohm
- R2 - 10-kilo-ohm
- R3, R4 - 100-ohm
- R5 - 4.7-kilo-ohm
- LDR1 - Light-dependent resistor

Miscellaneous:

- S1 - Push-to-on switch
- CON1 - 2-pin connector
- 6V battery /6V adaptor

Test Points

Test point	Details
TP0	0V, GND
TP1	6V
TP2	6V (when S1 pressed)
LED1	Blinks when a visitor is detected

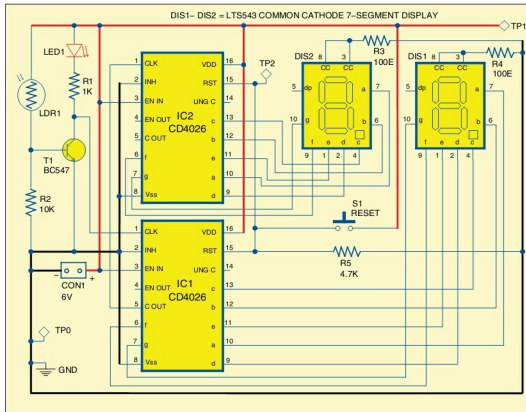


Fig. 1: Circuit of visitor counter

ohms in presence of bright light.

In this circuit, the amount of light falling on LDR1 decreases as a person crosses the entrance/exit gate and his shadow falls on LDR1. Consequently, the resistance of LDR1 increases to provide a clock pulse to pin 1 of IC1 through transistor T1. During this time, LED1 stops glowing momentarily, indicating that someone is entering or exiting the hall.

IC1 consists of a Johnson decade counter and an output decoder that converts the Johnson code into a 7-segment decoded output

for driving one stage in a numerical display. When it receives a clock at pin 1, it advances the count on display DIS1 by 'one.' Similarly, the count on the display advances by 'one' with each person entering through the gate. When the count reaches '9,' one cycle completes.

Carry-out pin 5 of IC1 is connected to clock pin 1 of IC2 to cascade another counter. On the next clock after count '9,' it goes high to provide a clock pulse to IC2, advancing its counter by one. Now IC1 starts all over again.

DIS1 shows the